Two Lifted Actions on Oriented Configuration Spaces: Stabilization and Bredon Homology

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Abstract

Let C_2 act on $M = \mathbb{R}^r \oplus \sigma$, $r \geq 2$, by reflection in the sign-representation coordinate. The action on unordered configurations has two lifts to the orientation double cover. Adding a free orbit exchanges these lifts. We combine them into a $\mathbb{Z}/2$-graded family with commuting equivariant stabilizations u and v, which add a fixed point and a free orbit, respectively. We determine its fixed-point branches, including the exceptional cases with zero or one fixed point, and describe both stabilization maps on each branch. Using ordinary homological stability, geometric fixed points, and isotropy separation, we prove that integral Bredon homology is finitely generated over $\mathbb{Z}[u, v]$ in every degree, and over $\mathbb{Z}[u, v^2]$ in each lift sector. We also compute the complete bigraded $\mathbb{Z}[u, v]$-module in degree zero. The presentation contains a class δ satisfying $2\delta = u^2\delta = 0$, while $v^k\delta$ and $uv^k\delta$ are nonzero for all $k \geq 0$. Thus the map induced by fixed-point stabilization on degree-zero Bredon homology fails to be injective in arbitrarily large arities in each lift sector.

Keywords: oriented configuration spaces; equivariant homological stability; Bredon homology; involutions; fixed points.

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1 Introduction

For $n \geq 2$, the oriented configuration space $C_n^+(M) = \mathrm{Conf}_n(M)/A_n$ is the double cover of the unordered configuration space $C_n(M) = \mathrm{Conf}_n(M)/\Sigma_n$ that retains an ordering modulo even permutations. Its orientation records the parity of an ordering of the points, independently of any orientation of M. We study how the two lifts of a reflection action of C_2 affect stabilization. Adding a free orbit exchanges the lifts, while adding fixed points can merge distinct fixed-point components. These effects lead us to consider both lifts together and to compare finite generation with the behavior of the individual stabilization maps.

Stabilization near an end of a manifold is defined by compressing the manifold and inserting one new point outside the image. Segal's end-stabilization theorem [13, Appendix to Section 5, Proposition A.1] and McDuff's scanning comparison [5, Theorem 1.2] are classical sources for the stable homology of unordered configurations on open manifolds. The integral ranges used below are Randal–Williams's range in the unordered case [9, Theorem A] and Palmer's range in the oriented case [8, Main Theorem].

Miller and Palmer identify the stable scanning target for oriented configurations as an explicit double cover of a section space [6, Theorem A and Section 3.6]. They also note that their end-stabilization map has two lifts between the oriented configuration covers, differing by the deck involution [6, Section 3.4]. This lift ambiguity concerns the stabilization map; the two lifts considered below instead lift the external C_2-action itself.

Equivariant scanning models for invariant labelled configurations were developed by Rourke and Sanderson [12, Theorems 1 and 5]. Belmont, Quigley, and Vogeli develop a Bredon homological finite-generation framework for unordered configurations on G-manifolds [1].

Let $C_2 = \{1, g\}$ act on a manifold M. The coordinatewise action of g on ordered configurations commutes with the alternating group and hence induces an involution on the oriented configuration space. Its composite with the deck involution is a second involution. These are the two lifts of the same action on the unordered configuration space.

Here

$$M = \mathbb{R}^r \oplus \sigma = \mathbb{R}^r \times \mathbb{R}, \qquad r \geq 2,$$

with $g(x,t) = (x,-t)$, and

$$F = M^{C_2} = \mathbb{R}^r \times \{0\} \cong \mathbb{R}^r.$$

For $\epsilon \in \mathbb{Z}/2$, write $C_n^{+,\epsilon}(M)$ for the oriented configuration space equipped with the ϵ-th lift. The two lifts are collected in the graded family

$$\mathcal{C}_{\mathrm{or}}(M) = \coprod_{\epsilon \in \mathbb{Z}/2} \coprod_{n \geq 2} C_n^{+,\epsilon}(M). \tag{1.1}$$

Adding a fixed point preserves the lift, whereas adding a free orbit interchanges the lifts. Writing ν for the deck involution, free-orbit stabilization satisfies $gv = \nu v g$; replacing v by its other lift νv does not change this exchange. The two-lift family is therefore needed to make free-orbit stabilization equivariant. The corresponding maps can be chosen to commute strictly:

$$u\colon C_n^{+,\epsilon}(M) \longrightarrow C_{n+1}^{+,\epsilon}(M), \qquad v\colon C_n^{+,\epsilon}(M) \longrightarrow C_{n+2}^{+,\epsilon+1}(M).$$

Their bidegrees are

$$|u| = (1,0), \qquad |v| = (2,1).$$

Thus the homology groups considered below form modules over

$$R = \mathbb{Z}[u,v],$$

graded by the number of points and by the lift index.

Every invariant unordered configuration is a finite C_2-set of the form

$$a[C_2/C_2] \amalg b[C_2/e].$$

The nontrivial element acts on its points as a product of b transpositions and therefore has sign $(-1)^b$. The lift indexed by ϵ has fixed points over this orbit-type stratum if and only if

$$b \equiv \epsilon \pmod 2.$$

We call this congruence the orientation-parity obstruction. It is compatible with free-orbit stabilization: v replaces b by $b+1$ and moves to the other lift.

For an allowed pair (a, b), let $D_{a,b}$ denote the corresponding fixed-point branch. The quotient of the free part of M is

$$B = (M - F)/C_2 \cong \mathbb{R}^r \times (0, \infty).$$

The fixed-point calculation gives

$$D_{a,b} \cong \begin{cases} C_a^+(F) \times C_b(B), & a \geq 2, \\ F \times C_b(B) \times \{\pm 1\}, & a = 1, \\ C_b(B) \times \{\pm 1\}, & a = 0. \end{cases} \tag{1.2}$$

The factors $\{\pm 1\}$ in the exceptional cases $a = 0, 1$ record an orientation bit that is absorbed into the orientation of the fixed-point configuration at the transition to $a = 2$. On branch indices the stabilization maps act by

$$u\colon (a, b) \longmapsto (a+1, b), \qquad v\colon (a, b) \longmapsto (a, b+1).$$

After forgetting the C_2-action, u is the usual oriented end stabilization. Palmer's theorem gives an isomorphism on $H_d(C_n^+(M); \mathbb{Z})$ for $n \geq 3d + 5$; see Theorem 6.1. Thus ordinary integral homological stability holds for the underlying oriented configuration spaces.

The ordinary homology of the fixed-point spaces is a separate input. The decomposition (1.2) identifies the two stabilization directions on each branch with familiar nonequivariant maps. In degree d, Palmer's theorem gives stability in the a-direction for $a \geq 3d + 5$, uniformly in b, while Randal–Williams's theorem gives stability in the b-direction for $b \geq 2d$, uniformly in a. The uniform statements follow from the natural integral Künneth sequence, including its Tor-terms. Consequently, the ordinary homology of the total fixed-point space is finitely generated over R. The equivariant argument uses this result together with finite generation of the underlying homology.

Bredon homology is taken at the C_2/C_2-level with coefficients in the constant Mackey functor $\underline{\mathbb{Z}}$; the conventions are specified in Section 7.

Even in degree zero, Bredon homology records fixed-point component data that ordinary homology of the underlying space does not detect. On the exceptional branches $a = 0, 1$, free-orbit stabilization preserves the two orientation components. Adding a second fixed point sends their images into the same path component. As shown in Section 8, this produces nonzero kernel classes for the map induced by u on degree-zero Bredon homology in arbitrarily large arities in each lift sector. For unordered configurations in the same reflection model, the corresponding map is injective in degree zero.

Despite these recurring kernels, the two stabilizations together generate Bredon homology from finitely many classes in each homological degree. This is the content of our main theorem.

Theorem 1.1 (Main theorem)**.** *For every $d \geq 0$, the module*

$$H_d^{C_2}(\mathcal{C}_{\mathrm{or}}(M); \underline{\mathbb{Z}})$$

is finitely generated over $\mathbb{Z}[u, v]$.

We use the geometric-fixed-point strategy of Belmont, Quigley, and Vogeli for unordered configurations [1]. The additional work concerns the orientation cover: we determine the fixed-point branches of the two-lift family in (1.2), including the two orientation components when $a = 0, 1$, and describe how both stabilizations act on them. For C_2, the isotropy-separation step is written out directly: the proper-isotropy term is controlled by the homotopy-orbit spectral sequence, and naturality makes all maps R-linear.

Set

$$\mathcal{C}_{\mathrm{or}}^{\epsilon}(M) = \coprod_{n \geq 2} C_n^{+,\epsilon}(M).$$

The map v interchanges the two sectors, while u and v^2 preserve each one. The ring $\mathbb{Z}[u, v]$ is finite free over $\mathbb{Z}[u, v^2]$, and the two sectors give a direct-sum decomposition of Bredon homology over the latter ring. Thus Theorem 1.1 gives the following sectorwise consequence.

Corollary 1.2. *For every $d \geq 0$ and $\epsilon \in \mathbb{Z}/2$,*

$$H_d^{C_2}(\mathcal{C}_{\mathrm{or}}^{\epsilon}(M); \underline{\mathbb{Z}})$$

is finitely generated over $\mathbb{Z}[u, v^2]$.

Degree zero admits a more precise description. Let x be the class of $D_{2,0}$, let p be one of the two component classes of $D_{0,1}$, and let δ be the difference of those two component classes.

Theorem 1.3 (Degree-zero presentation)**.** *There is an isomorphism of bigraded R-modules*

$$H_0^{C_2}(\mathcal{C}_{\mathrm{or}}(M); \underline{\mathbb{Z}}) \cong \frac{R\langle x, p\rangle}{(vx - u^2 p,\ 2(vp - u^2 x))} \oplus \frac{R}{(2, u^2)}\langle \delta \rangle. \tag{1.3}$$

Here $|x| = (2, 0)$ and $|p| = |\delta| = (2, 1)$.

The proof is given in Theorem 8.3.

The relation $vx = u^2 p$ records an actual coincidence of stabilized fixed components. The class $vp - u^2 x$ records two distinct fixed components of $C_4^{+,0}(M)$, whose underlying space is connected, and hence has order two. The summand $R/(2, u^2)\langle \delta \rangle$ records the exceptional orientation components: $v^k \delta$ and $uv^k \delta$ are nonzero for every $k \geq 0$, while $u^2 v^k \delta = 0$. The classes $uv^k \delta$, in arity $3 + 2k$ and lift $\overline{k+1}$, are the nonzero kernel classes described above.

Related work. A complementary organization of all arities is given by Randal–Williams: after one-point compactification, the unordered configuration spaces form an arity-graded commutative monoid under superposition, with a derived presentation [10, Theorem 1.1].[1]

For comparison, let G be a finite group and V a real G-representation. When $\dim V \geq 2$ and V contains a one-dimensional trivial summand, Dugger and Hazel give a presentation of the $RO(G)$-graded Bredon cohomology ring of the ordered configuration space $\mathrm{OC}_q(V)$ at fixed arity [3, Theorem 5.2]. In our notation, $\mathrm{OC}_q(V) = \mathrm{Conf}_q(V)$. Their calculation concerns the ordered

[1] The result recalled here concerns unlabelled configurations. The correction in [7, Section 1] concerns the positive-rank extension in [10, Theorem 1.2].

space rather than its A_q-quotient, and does not involve the orientation double cover or its two lifted actions.

Sections 2–5 construct the two lifts, identify their fixed points, and describe stabilization on each branch. Section 6 proves ordinary finite generation, and Section 7 deduces the main theorem and its sectorwise corollary. Section 8 gives the degree-zero presentation. We do not seek sharp bounds on the arities of generators.

2 Oriented configurations and the two lifted actions

Convention 2.1. Fix $r \geq 2$. Let σ denote the nontrivial one-dimensional real representation of $C_2 = \{1, g\}$, and put

$$M = \mathbb{R}^r \oplus \sigma = \mathbb{R}^r \times \mathbb{R}, \qquad g(x,t) = (x,-t).$$

The fixed subspace is

$$F = M^{C_2} = \mathbb{R}^r \times \{0\}.$$

All configuration spaces in the oriented family have arity $n \geq 2$.

For a Hausdorff space X, define the ordered configuration space

$$\mathrm{Conf}_n(X) = \{(x_1, \ldots, x_n) \in X^n : x_i \neq x_j \text{ for } i \neq j\}.$$

The symmetric group Σ_n acts freely by permuting coordinates. We write

$$C_n(X) = \mathrm{Conf}_n(X)/\Sigma_n, \qquad C_n^+(X) = \mathrm{Conf}_n(X)/A_n.$$

For ordinary configuration spaces we use the standard convention $C_0(X) = *$. $C_n^+(X)$ remembers an ordering up to even permutation. A point will be written (Q, o), where $Q \subset X$ is an unordered n-point subset and o is one of its two orientations. There is a two-sheeted covering

$$p_n \colon C_n^+(X) \longrightarrow C_n(X). \tag{2.1}$$

For a connected manifold N of dimension at least two, $\mathrm{Conf}_n(N)$ is path connected. Indeed, the complement of any finite subset of N is path connected. Given two ordered configurations, choose a third whose points are disjoint from both. Moving one point at a time, while avoiding all the remaining points, connects each given configuration to the third. It follows that $C_n(N)$ and $C_n^+(N)$ are path connected as well.

Definition 2.2. The *deck involution*

$$\nu \colon C_n^+(X) \longrightarrow C_n^+(X)$$

is induced by any odd permutation. Equivalently,

$$\nu(Q, o) = (Q, -o).$$

The coordinatewise action of g on $\mathrm{Conf}_n(M)$ commutes with the entire symmetric-group action. It therefore descends both to $C_n(M)$ and to $C_n^+(M)$, and on the latter it commutes strictly with ν.

Definition 2.3 (The two lifts)**.** For $\epsilon \in \mathbb{Z}/2$, define

$$g_\epsilon = \nu^\epsilon g \colon C_n^+(M) \longrightarrow C_n^+(M).$$

The resulting C_2-space is denoted $C_n^{+,\epsilon}(M)$.

Proposition 2.4. *For each $\epsilon \in \mathbb{Z}/2$, the map g_ϵ is an involution and lifts the given action on $C_n(M)$. The two actions g_0 and g_1 are the two lifts through the double cover* (2.1).

Proof. Since $g\nu = \nu g$ and both maps square to the identity,

$$g_\epsilon^2 = \nu^{2\epsilon} g^2 = 1.$$

Since $p_n \nu = p_n$, both g_0 and g_1 cover the coordinatewise action on $C_n(M)$. Any lift of this action takes each $z \in C_n^+(M)$ to either $g(z)$ or $\nu g(z)$. This choice is locally constant by the covering property and hence constant because $C_n^+(M)$ is connected. Thus g_0 and g_1 are exactly the two lifts. □

Definition 2.5 (The graded family)**.** Set

$$\mathcal{C}_{\mathrm{or}}(M) = \coprod_{\substack{n \geq 2 \\ \epsilon \in \mathbb{Z}/2}} C_n^{+,\epsilon}(M),$$

with the disjoint-union topology and the componentwise C_2-action. We regard ϵ as an external $\mathbb{Z}/2$-grading.

The two copies remain distinct: they have the same underlying oriented configuration space but different C_2-actions. The free-orbit stabilization in Section 4 moves from one copy to the other.

Remark 2.6 (The arity truncation). The arity cutoff ensures that p_n is a double cover. The absent pairs $(a, b) = (0, 0)$ and $(1, 0)$ will be accounted for explicitly when the finite generating region is chosen in Proposition 6.6.

3 Fixed points and the orientation-parity obstruction

Let $Q \in C_n(M)$ be fixed by the coordinatewise C_2-action. As a finite C_2-set it has a unique decomposition

$$Q \cong a[C_2/C_2] \amalg b[C_2/e], \qquad n = a + 2b, \tag{3.1}$$

for integers $a, b \geq 0$. The first summand consists of the points of $Q \cap F$; the second consists of the free two-point orbits.

Proposition 3.1 (Orientation parity)**.** *Let $(Q, o) \in C_n^{+,\epsilon}(M)$ lie over an invariant unordered configuration of orbit type (a, b). Then*

$$g_\epsilon(Q, o) = (Q, (-1)^{b+\epsilon} o).$$

Consequently (Q, o) is C_2-fixed if and only if

$$b \equiv \epsilon \pmod 2. \tag{3.2}$$

When (3.2) *holds, both orientations over Q are fixed.*

Proof. The action of g on the points of Q fixes the a points in F and transposes the two points in each of the b free orbits. Its sign is therefore $(-1)^b$. Thus the coordinatewise lift sends o to $(-1)^b o$, and the additional factor ν^ϵ contributes $(-1)^\epsilon$. This gives the displayed formula and the criterion. □

Definition 3.2 (Fixed-point branches)**.** For $a, b \geq 0$ with $a + 2b \geq 2$, let

$$D_{a,b} \subset \left(C^{+,\bar{b}}_{a+2b}(M)\right)^{C_2}$$

be the subspace lying over invariant unordered configurations of orbit type (a, b), where $\bar{b}$ is the class of b in $\mathbb{Z}/2$.

The branches are open and closed in the fixed-point space. To see this, choose pairwise disjoint balls about the points of an invariant configuration, with the balls permuted by g. A sufficiently nearby configuration has exactly one point in each ball. If it is invariant, its point in a ball centred on F must be fixed, while points in the remaining balls occur in free pairs. Thus a and b are locally constant among invariant configurations. Proposition 3.1 gives

$$\mathcal{C}_{\mathrm{or}}(M)^{C_2} = \coprod_{\substack{a,b\geq 0 \\ a+2b\geq 2}} D_{a,b}. \tag{3.3}$$

We now identify every branch. Write

$$M_{(e)} = M - F = \mathbb{R}^r \times (\mathbb{R} \setminus \{0\})$$

for the free locus. The quotient is

$$B = M_{(e)}/C_2 \cong M_{>0} := \mathbb{R}^r \times (0, \infty). \tag{3.4}$$

The positive half-space is a global section of the double cover $M_{(e)} \to B$; in particular, this cover is trivial.

Let $I_{a,b}$ denote the orbit-type (a, b) stratum in $C_{a+2b}(M)^{C_2}$. Sending a configuration to its fixed points in F and its free orbits in B gives a homeomorphism

$$I_{a,b} \cong C_a(F) \times C_b(B). \tag{3.5}$$

Lemma 3.3 (Orientation monodromy on an orbit-type stratum)**.** *Under* (3.5)*, the restriction of the orientation cover*

$$C^+_{a+2b}(M) \longrightarrow C_{a+2b}(M)$$

to $I_{a,b}$ has monodromy equal to the sign of the permutation of the a fixed points. Permuting free two-point blocks contributes no sign.

Proof. Use the section $M_{>0} \to M_{(e)}$ to write each free orbit uniquely as the ordered pair (y, gy) with $y \in M_{>0}$. Along a loop in $C_a(F) \times C_b(B)$, let $\pi \in \Sigma_a$ be the induced permutation of fixed points and $\tau \in \Sigma_b$ the induced permutation of free blocks. The induced permutation of all $a + 2b$ points has sign

$$\mathrm{sgn}(\pi)\,\mathrm{sgn}(\tau)^2 = \mathrm{sgn}(\pi).$$

There is no additional transposition inside a free block: the chosen positive lift varies continuously and returns to the positive lift because the free-locus cover has the global section $M_{>0}$. Pure braiding contributes no permutation to the fiber of the orientation cover. Hence the monodromy is precisely the stated sign character. □

Theorem 3.4 (Fixed-point decomposition)**.** *For every $a, b \geq 0$ with $a+2b \geq 2$, the positive-half-space choice in* (3.4) *gives homeomorphisms*

$$D_{a,b} \cong \begin{cases} C_a^+(F) \times C_b(B), & a \geq 2, \\ F \times C_b(B) \times \{\pm 1\}, & a = 1, \\ C_b(B) \times \{\pm 1\}, & a = 0. \end{cases} \tag{3.6}$$

In particular, $D_{a,b}$ is connected for $a \geq 2$ and has exactly two path components for $a = 0, 1$.

Proof. When the parity condition holds, Proposition 3.1 shows that $D_{a,b}$ is the full restriction of the orientation double cover to $I_{a,b}$. By Lemma 3.3, for $a \geq 2$ this cover is pulled back from the usual orientation cover $C_a^+(F) \to C_a(F)$, giving the first line of (3.6).

For $a = 0$ or 1, no nontrivial permutation of the fixed points is possible. The monodromy is therefore trivial, so the restricted two-sheeted cover is the product with $\{\pm 1\}$. Since $C_1(F) = F$ and $C_0(F)$ is a point, this gives the remaining two lines. The factors F, B, $C_b(B)$, and $C_a^+(F)$ for $a \geq 2$ are path connected, which proves the component statement. □

The low-arity fixed loci provide useful checks on both the parity and the exceptional lines. Empty terms are omitted.

Example 3.5 (Arities two through five). Using Theorem 3.4, one obtains

$$\begin{aligned}
\big(C_2^{+,0}(M)\big)^{C_2} &\cong C_2^+(F), \\
\big(C_2^{+,1}(M)\big)^{C_2} &\cong B \times \{\pm 1\}, \\
\big(C_3^{+,0}(M)\big)^{C_2} &\cong C_3^+(F), \\
\big(C_3^{+,1}(M)\big)^{C_2} &\cong F \times B \times \{\pm 1\}, \\
\big(C_4^{+,0}(M)\big)^{C_2} &\cong C_4^+(F) \amalg C_2(B) \times \{\pm 1\}, \\
\big(C_4^{+,1}(M)\big)^{C_2} &\cong C_2^+(F) \times B, \\
\big(C_5^{+,0}(M)\big)^{C_2} &\cong C_5^+(F) \amalg F \times C_2(B) \times \{\pm 1\}, \\
\big(C_5^{+,1}(M)\big)^{C_2} &\cong C_3^+(F) \times B.
\end{aligned}$$

Remark 3.6. The parity obstruction is not a statement that invariant configurations fail to lift to the orientation cover. Every invariant unordered configuration has two lifts as a point of the double cover. Rather, it specifies under which of the two lifted C_2-actions those points are fixed.

4 Stabilization maps and the bigraded system

We choose explicit compression embeddings so that the two stabilizations commute strictly. Let

$$h(s) = 1 - e^{-s} \colon \mathbb{R} \longrightarrow (-\infty, 1)$$

and define

$$\begin{aligned}
e_u(x_1, x_2, \ldots, x_r, t) &= (h(x_1), x_2, \ldots, x_r, t), \\
e_v(x_1, \ldots, x_r, t) &= (x_1, \ldots, x_r, \tanh t).
\end{aligned}$$

Both are C_2-equivariant embeddings. Choose

$$q = (2, 0, \dots, 0, 0) \in F, \qquad p = (0, \dots, 0, 2), \qquad gp = (0, \dots, 0, -2).$$

Then $q \notin e_u(M)$, while $p, gp \notin e_v(M)$. Moreover,

$$e_u e_v = e_v e_u, \qquad e_u(p) = p, \quad e_u(gp) = gp, \quad e_v(q) = q. \tag{4.1}$$

On ordered configurations define

$$\widetilde{u}(x_1, \dots, x_n) = (e_u x_1, \dots, e_u x_n, q),$$
$$\widetilde{v}(x_1, \dots, x_n) = (e_v x_1, \dots, e_v x_n, p, gp).$$

These maps are equivariant for the standard inclusions $A_n \hookrightarrow A_{n+1}$ and $A_n \hookrightarrow A_{n+2}$, and hence descend to maps of oriented configuration spaces

$$u\colon C_n^+(M) \longrightarrow C_{n+1}^+(M), \tag{4.2}$$
$$v\colon C_n^+(M) \longrightarrow C_{n+2}^+(M). \tag{4.3}$$

Lemma 4.1 (Relations with the deck involution and the coordinatewise action)**.** *On oriented configuration spaces,*

$$u\nu = \nu u, \qquad gu = ug,$$
$$v\nu = \nu v, \qquad gv = \nu v g.$$

Proof. The maps $\widetilde{u}$ and $\widetilde{v}$ apply the same embedding to all old coordinates, so they commute with odd as well as even permutations of those coordinates. This proves the two relations with ν.

The point q is fixed and e_u is equivariant, so applying g before or after u gives the same oriented configuration. For v, applying g changes the appended ordered block (p, gp) to (gp, p). The two blocks differ by one transposition, which is precisely the deck involution. Thus $gv = \nu v g$. □

Proposition 4.2 (Equivariant stabilizations)**.** *For every $n \geq 2$ and $\epsilon \in \mathbb{Z}/2$, the maps* (4.2) *and* (4.3) *define C_2-equivariant maps*

$$u\colon C_n^{+,\epsilon}(M) \longrightarrow C_{n+1}^{+,\epsilon}(M),$$
$$v\colon C_n^{+,\epsilon}(M) \longrightarrow C_{n+2}^{+,\epsilon+1}(M).$$

Proof. Using Lemma 4.1 and the commutation of u, v with ν,

$$g_\epsilon u = \nu^\epsilon g u = \nu^\epsilon u g = u g_\epsilon.$$

For the free-orbit stabilization,

$$g_{\epsilon+1} v = \nu^{\epsilon+1} g v = \nu^{\epsilon+1} \nu v g = v \nu^\epsilon g = v g_\epsilon.$$

These are exactly the required equivariance identities. □

Replacing v by its other lift νv does not remove the exchange of lift sectors: both choices satisfy

$$g_\epsilon w = \nu w g_\epsilon \qquad \text{for } w = v \text{ or } \nu v.$$

Thus reversing the orientation of the added free orbit does not make this stabilization equivariant within a single lift sector.

Proposition 4.3 (Strict commutation)**.** *The endomorphisms u and v of the total family $\mathcal{C}_{\mathrm{or}}(M)$ satisfy*

$$uv = vu$$

as maps, not merely up to homotopy.

Proof. By (4.1), the old configuration points agree in the two composites. The composite uv appends the ordered block (p, gp, q), while vu appends (q, p, gp). Passing from the first list to the second moves q across two entries and is therefore an even permutation. The two ordered lists define the same point of the quotient by A_{n+3}. □

Definition 4.4 (Polynomial action)**.** Give a component $C_n^{+,\epsilon}(M)$ bidegree (n, ϵ). Then

$$|u| = (1, 0), \qquad |v| = (2, 1).$$

By Proposition 4.3, the induced maps on any functorial homology theory make the homology of $\mathcal{C}_{\mathrm{or}}(M)$ a bigraded module over

$$R = \mathbb{Z}[u, v].$$

Remark 4.5. The relation $gv = \nu vg$ is the stabilization form of the parity obstruction. Appending a free orbit adds one transposition to the permutation induced by g, so equivariance forces the target lift to change.

5 Fixed-point branches and the action of stabilization

We now make (3.6) compatible with the maps u and v. Under the identification $B \cong M_{>0}$ of (3.4), the two compression embeddings induce

$$\begin{aligned} \bar{e}_u(x_1, x_2, \ldots, x_r, t) &= (h(x_1), x_2, \ldots, x_r, t), \\ \bar{e}_v(x_1, \ldots, x_r, t) &= (x_1, \ldots, x_r, \tanh t). \end{aligned}$$

Let $\bar{p} \in B$ be the free orbit represented by $p \in M_{>0}$. Define

$$t_B \colon C_b(B) \longrightarrow C_{b+1}(B), \qquad Z \longmapsto C_b(\bar{e}_v)(Z) \cup \{\bar{p}\}. \tag{5.1}$$

On the fixed subspace, let $e_u^F = e_u|_F$, and for $a \geq 2$ define the oriented stabilization

$$t_F^+ \colon C_a^+(F) \longrightarrow C_{a+1}^+(F), \qquad [x_1, \ldots, x_a] \longmapsto [e_u^F x_1, \ldots, e_u^F x_a, q]. \tag{5.2}$$

To fix the signs on the exceptional branches, we use the positive section of $M_{(e)} \to B$ and order all fixed points before all free pairs. On $D_{0,b}$, the sign $+$ is represented by an ordered list

$$(y_1, gy_1, \ldots, y_b, gy_b), \qquad y_i \in M_{>0},$$

and $-$ is obtained by applying ν. On $D_{1,b}$, the fixed point is placed first, followed by the same free blocks. For $a \geq 2$, an orientation of the fixed points is followed by the free blocks. Changing the order of the blocks is even and hence does not affect these conventions.

Proposition 5.1 (Action on branch indices)**.** *The stabilizations restrict to maps of fixed-point branches*

$$u \colon D_{a,b} \longrightarrow D_{a+1,b}, \qquad v \colon D_{a,b} \longrightarrow D_{a,b+1}.$$

Proof. The map u appends the fixed point q, and hence adds one orbit of type C_2/C_2. The map v appends $\{p, gp\}$, and hence adds one orbit of type C_2/e. Proposition 4.2 puts both images in the lift required by the parity condition. □

Proposition 5.2 (The free-orbit stabilization)**.** *Under the homeomorphisms of Theorem 3.4, the map v is*

$$\begin{cases} \mathrm{id}_{C_a^+(F)} \times t_B, & a \geq 2, \\ \mathrm{id}_F \times t_B \times \mathrm{id}_{\{\pm 1\}}, & a = 1, \\ t_B \times \mathrm{id}_{\{\pm 1\}}, & a = 0. \end{cases}$$

These formulas are equalities for the chosen identifications, not only homotopies.

Proof. The embedding e_v fixes F pointwise. On free orbits it induces $\bar{e}_v$ and appends $\bar{p}$, which is exactly t_B. The appended ordered pair is (p, gp), the positive-section convention for the new free block, so the orientation bit is unchanged on the lines $a = 0, 1$. □

The fixed-point stabilization has one transition that absorbs the orientation bit. Define

$$\lambda_F \colon F \times \{\pm 1\} \longrightarrow C_2^+(F) \tag{5.3}$$

by

$$\lambda_F(x, +) = [e_u^F(x), q], \qquad \lambda_F(x, -) = [q, e_u^F(x)] = \nu[e_u^F(x), q].$$

Proposition 5.3 (The fixed-point stabilization)**.** *Under the homeomorphisms of Theorem 3.4, the map u is as follows.*

1. *If $a \geq 2$, then*
$$u = t_F^+ \times C_b(\bar{e}_u).$$
2. *From $a = 0$ to $a = 1$,*
$$u(Z, \eta) = (q, C_b(\bar{e}_u)(Z), \eta).$$
3. *From $a = 1$ to $a = 2$,*
$$u(x, Z, \eta) = \big(\lambda_F(x, \eta), C_b(\bar{e}_u)(Z)\big).$$

In particular, the orientation bit is preserved by the first application of u and becomes the orientation of the two-point fixed configuration after the second.

Proof. On every old point, u applies e_u; on the free-orbit quotient this is $\bar{e}_u$. In the ordered definition, the new fixed point q appears after all old points. Moving it left across the $2b$ free points is an even permutation, so it does not change the orientation. If $a \geq 2$, this gives t_F^+ on the oriented fixed-point factor. If $a = 0$, the same even move places q before the free blocks without changing η. If $a = 1$, the two choices of η give the two orientations $[e_u^F(x), q]$ and $[q, e_u^F(x)]$, which is precisely λ_F. □

The self-embedding in the B-factor of u can be removed by a homotopy.

Lemma 5.4. *The embedding $\bar{e}_u \colon B \to B$ is isotopic through embeddings to the identity. Consequently $C_b(\bar{e}_u)$ is homotopic to the identity of $C_b(B)$ for every b.*

Proof. For $s \in [0,1]$, set

$$h_s(x) = (1-s)(1-e^{-x}) + sx.$$

Its derivative is

$$h_s'(x) = (1-s)e^{-x} + s > 0,$$

so h_s is an embedding for every s. Applying h_s in the first coordinate and fixing all other coordinates gives the required isotopy from $\bar{e}_u$ to the identity. Applying configurations functorially gives the final statement. □

Corollary 5.5. *For $a \geq 2$, the branch map*

$$u\colon D_{a,b} \longrightarrow D_{a+1,b}$$

is homotopic to $t_F^+ \times \mathrm{id}_{C_b(B)}$.

Remark 5.6. For $a = 0, 1$, each branch $D_{a,b}$ has two components. At each $x \in F$, the two labels on $D_{1,b}$ map to the two orientations of $\{e_u^F(x), q\}$. These are distinct points of $C_2^+(F)$, but they lie in the same path component.

6 Ordinary homological stability and finite generation

We first record the two external stability theorems in the precise forms used below.

Theorem 6.1 (Palmer)**.** *Let N be the interior of a connected manifold with nonempty boundary, of dimension at least two. The oriented stabilization*

$$C_a^+(N) \longrightarrow C_{a+1}^+(N)$$

is an isomorphism on integral homology in degree i when

$$i \leq \frac{a-5}{3}.$$

Source and comparison of stabilization maps.
This is the integral isomorphism range in Palmer's theorem [8, Main Theorem]. Palmer's stabilization is obtained by compressing toward a chosen boundary component and appending a point. For $N = F$ or M, write $N = \mathbb{R}^m$, with $m = r$ or $r+1$. Adjoining the positive endpoint of the first coordinate gives a manifold with boundary $(-\infty, +\infty] \times \mathbb{R}^{m-1}$ whose interior is N; Palmer's theorem does not require this manifold to be compact. Choose a smooth increasing collar compression $k\colon \mathbb{R} \to (-\infty, 2)$ that equals the identity for sufficiently negative arguments and sends the adjoined endpoint to 2. Applied to the first coordinate and followed by appending $q = (2, 0, \ldots, 0)$, this is Palmer's end stabilization.

The compression $h(s) = 1 - e^{-s}$ used for t_F^+ and u is joined to k by $h_\lambda = (1-\lambda)h + \lambda k$, $0 \leq \lambda \leq 1$. Each h_λ is a strictly increasing embedding with values less than 2. Applying it to the first coordinate therefore gives an isotopy of embeddings whose images avoid q. Appending q last throughout this isotopy gives a homotopy on oriented configuration spaces. Thus t_F^+ and the underlying map u have the required stabilization homotopy class; independence of the collar choice is also recorded in [8, Remark 2.4]. □

Theorem 6.2 (Randal–Williams)**.** *Let N be a connected open manifold that is the interior of a compact manifold with nonempty boundary, with* $\dim N > 1$*. The unordered stabilization*

$$C_b(N) \longrightarrow C_{b+1}(N)$$

is an integral homology isomorphism in degree i when

$$2i \leq b.$$

Source and comparison of stabilization maps.
This is the isomorphism assertion of Randal–Williams's theorem [9, Theorem A]; the construction of the stabilization is given in [9, Section 4]. The space $B \cong \mathbb{R}^r \times (0,\infty) \cong \mathbb{R}^{r+1}$ is the interior of a closed ball. To compare t_B with the cited stabilization, use the coordinate $y = 1/t$. Then $\bar{p} = (0, 1/2)$ and $\bar{e}_v(x, y) = (x, \coth(1/y))$.

Choose smooth compactly supported functions $\beta\colon \mathbb{R}^r \to [0,1]$ and $\chi\colon [0,\infty) \to [0,1]$, with $\beta(0) = 1$, $\chi = 1$ near zero, and $\chi' > -2$. Put

$$k_x(y) = y + \tfrac{1}{2}\beta(x)\chi(y).$$

Since $\partial_y k_x > 0$, the map $(x, y) \mapsto (x, k_x(y))$ embeds the closed half-space into itself, sends its interior into itself, and takes the boundary origin to $\bar{p}$. It is the identity outside a compact boundary neighborhood and is isotopic to the identity by multiplying the added term by a parameter in $[0, 1]$. Thus it extends by the identity over the boundary point at infinity in the closed-ball compactification of the half-space. After rescaling the boundary chart to place the support inside its unit ball, this is the local boundary compression used in the cited construction.

For $0 \leq \lambda \leq 1$, define on the open half-space

$$K_\lambda(x, y) = \big(x, (1-\lambda)\coth(1/y) + \lambda k_x(y)\big).$$

The last coordinate is positive and strictly increasing in y, so every K_λ is an embedding. Its image avoids $\bar{p}$: at $x = 0$, one has $\coth(1/y) > 1$ and $k_0(y) > k_0(0) = 1/2$ for $y > 0$. Applying K_λ to each configuration point and adjoining $\bar{p}$ therefore gives a homotopy from t_B to this standard stabilization. The stated integral stability range consequently applies to t_B. □

Lemma 6.3 (Finite type)**.** *For fixed $n \geq 2$ and $a, b \geq 0$ with $a + 2b \geq 2$, the integral homology groups of $C_n^+(M)$, $C_b(B)$, and $D_{a,b}$ are finitely generated abelian groups in every degree. The same holds for $C_a^+(F)$ when $a \geq 2$.*

Proof. The spaces M, F, and B are semialgebraic, and their ordered configuration spaces are obtained from finite Cartesian powers by removing finite unions of linear diagonals. For a finite group G acting by coordinate permutations on an ambient Euclidean space, the coefficients of

$$\prod_{\gamma \in G} (1 + \langle \mathbf{t}, \gamma x \rangle),$$

together with $\|x\|^2$, define an invariant polynomial map that separates G-orbits. For nonzero x, all the displayed linear factors are nonconstant and have constant term one. Unique factorization in $\mathbb{R}[\mathbf{t}]$ therefore recovers the orbit multiset from the product; the zero orbit is distinguished by the norm coordinate. This coordinate also makes the map proper, so the induced map from the ambient orbit space is a homeomorphism onto its closed image. Restricting to a configuration locus identifies each quotient by A_n, A_a, or Σ_b with a semialgebraic set: apply the Tarski–Seidenberg projection

theorem [2, Theorem 2.3 and Corollary 2.4] to the graph of the polynomial map restricted to that locus.

The configuration locus is invariant and open, so each such quotient X is locally closed. Embed it semialgebraically into the open unit ball by radial contraction and let $\overline{X}$ be its closure; then $A = \overline{X} \setminus X$ is a closed semialgebraic subset of the compact semialgebraic set $\overline{X}$. The compatible semialgebraic triangulation theorem [2, Theorem 3.12] identifies $(\overline{X}, A)$ with a finite simplicial pair $(|K|, |L|)$. Hence $X \cong |K| \setminus |L|$. In the barycentric subdivision $\operatorname{sd} K$, renormalizing barycentric coordinates away from the vertices belonging to $\operatorname{sd} L$ retracts this complement onto the finite full subcomplex spanned by barycentres of simplices of K not in L. Thus X has finite CW type and finitely generated integral homology. The claim for $D_{a,b}$ follows from (3.6), since finite products and finite disjoint unions preserve finite CW type. □

The next two propositions are the ordinary stability input for the fixed-point space.

Proposition 6.4 (Uniform stability in the fixed-point direction)**.** *Fix $d \geq 0$. If*

$$a \geq N_d := 3d + 5,$$

then for every $b \geq 0$ the map

$$u_* \colon H_d(D_{a,b}; \mathbb{Z}) \longrightarrow H_d(D_{a+1,b}; \mathbb{Z})$$

is an isomorphism.

Proof. The inequality forces $a \geq 2$, so Corollary 5.5 identifies the map up to homotopy with

$$t_F^+ \times \mathrm{id}_{C_b(B)}\,.$$

For every $i \leq d$, the inequality $a \geq 3d + 5$ implies $i \leq (a-5)/3$. Thus $(t_F^+)_*$ is an isomorphism on $H_i(-; \mathbb{Z})$ by Theorem 6.1.

Apply the natural integral Künneth short exact sequence

$$\begin{aligned} 0 \longrightarrow \bigoplus_{i+j=d} H_i(X) \otimes H_j(Y) \longrightarrow H_d(X \times Y) \\ \longrightarrow \bigoplus_{i+j=d-1} \operatorname{Tor}_1^{\mathbb{Z}}(H_i(X), H_j(Y)) \longrightarrow 0 \end{aligned} \tag{6.1}$$

to $X = C_a^+(F)$, $Y = C_b(B)$, and to the corresponding target. The maps on both outer terms are isomorphisms, because only homological degrees at most d occur. Naturality of (6.1) and the short five lemma give the result. No torsion-freeness assumption is needed. □

Proposition 6.5 (Uniform stability in the free-orbit direction)**.** *Fix $d \geq 0$. If*

$$b \geq 2d,$$

then for every $a \geq 0$ for which $D_{a,b}$ exists, the map

$$v_* \colon H_d(D_{a,b}; \mathbb{Z}) \longrightarrow H_d(D_{a,b+1}; \mathbb{Z})$$

is an isomorphism.

Proof. Put

$$X_a = \begin{cases} C_a^+(F), & a \geq 2, \\ F \times \{\pm 1\}, & a = 1, \\ \{\pm 1\}, & a = 0. \end{cases}$$

Then $D_{a,b} \cong X_a \times C_b(B)$, and Proposition 5.2 identifies v with $\mathrm{id}_{X_a} \times t_B$. For every $j \leq d$, the assumption $b \geq 2d$ implies $2j \leq b$, so $(t_B)_*$ is an isomorphism on H_j by Theorem 6.2. Applying the natural Künneth sequence (6.1), including its Tor-terms, proves the claim exactly as in Proposition 6.4. □

We turn stability in the two coordinates into finite generation for the disjoint union. Recall that singular homology takes disjoint unions to direct sums, so

$$H_d(\mathcal{C}_{\mathrm{or}}(M)^{C_2}; \mathbb{Z}) \cong \bigoplus_{\substack{a,b \geq 0 \\ a+2b \geq 2}} H_d(D_{a,b}; \mathbb{Z}).$$

Proposition 6.6 (Fixed-point finite generation)**.** *For every $d \geq 0$, the R-module*

$$H_d(\mathcal{C}_{\mathrm{or}}(M)^{C_2}; \mathbb{Z})$$

is finitely generated.

Proof. Set

$$N_d = 3d + 5, \qquad B_d = \max\{2d, 1\}.$$

Consider the finite set of branch indices

$$\mathcal{R}_d = \{(a, b) : 0 \leq a \leq N_d,\ 0 \leq b \leq B_d,\ a + 2b \geq 2\}. \tag{6.2}$$

By Lemma 6.3, the direct sum of $H_d(D_{a,b}; \mathbb{Z})$ over $\mathcal{R}_d$ has a finite set of abelian-group generators.

Let $\alpha \in H_d(D_{a,b}; \mathbb{Z})$ be arbitrary. If $a > N_d$, repeated use of the isomorphisms in Proposition 6.4 expresses α as $u^{a-N_d}\alpha'$ for some $\alpha' \in H_d(D_{N_d,b}; \mathbb{Z})$. If $a \leq N_d$, leave the a-coordinate unchanged. Next, if $b > B_d$, repeated use of Proposition 6.5 expresses the resulting class as a power of v applied to a class with $b = B_d$. The two stabilizations commute, so α lies in the R-span of the homology of the finite region (6.2).

The lower bound $B_d \geq 1$ is needed only for the exceptional lines: $D_{0,0}$ and $D_{1,0}$ are absent because of the global arity cutoff $n \geq 2$. Thus every terminal branch used above actually belongs to $\mathcal{C}_{\mathrm{or}}(M)^{C_2}$. □

The underlying space has two copies of each oriented configuration space:

$$\mathcal{C}_{\mathrm{or}}(M)^e = \coprod_{\epsilon \in \mathbb{Z}/2} \coprod_{n \geq 2} C_n^+(M).$$

Proposition 6.7 (Underlying finite generation)**.** *For every $d \geq 0$, the R-module*

$$H_d(\mathcal{C}_{\mathrm{or}}(M)^e; \mathbb{Z})$$

is finitely generated.

Proof. On each lift sector, u is the oriented end stabilization of $M \cong \mathbb{R}^{r+1}$. By Theorem 6.1, it is an isomorphism on H_d for $n \geq 3d+5$. The homology groups in the finitely many arities $2 \leq n \leq 3d+5$ are finitely generated by Lemma 6.3. They therefore give a finite set of generators over $\mathbb{Z}[u]$ for both sectors. The same set generates over the larger ring R. □

We shall need ordinary homology with finitely generated coefficients after taking geometric fixed points.

Lemma 6.8 (Change of coefficients)**.** *Let Y carry commuting self-maps giving an action of the Noetherian ring $R = \mathbb{Z}[u, v]$. Suppose $H_j(Y;\mathbb{Z})$ is a finitely generated R-module for every j. If A is a finitely generated abelian group, then $H_j(Y;A)$ is a finitely generated R-module for every j.*

Proof. Write A as a finite direct sum of copies of $\mathbb{Z}$ and of cyclic groups $\mathbb{Z}/m$. For a cyclic summand, the natural universal-coefficient sequence is

$$0 \longrightarrow H_j(Y;\mathbb{Z}) \otimes \mathbb{Z}/m \longrightarrow H_j(Y;\mathbb{Z}/m) \longrightarrow \operatorname{Tor}_1^{\mathbb{Z}}\big(H_{j-1}(Y;\mathbb{Z}), \mathbb{Z}/m\big) \longrightarrow 0.$$

It is R-linear by naturality. The tensor term is a quotient of a finitely generated R-module. The Tor-term is the m-torsion submodule of $H_{j-1}(Y;\mathbb{Z})$; it is finitely generated because R is Noetherian. An extension of two finitely generated modules is finitely generated. Finite direct sums complete the proof. □

7 From geometric fixed points to Bredon homology

We use genuine C_2-spectra and the constant Mackey functor $\underline{\mathbb{Z}}$. Its value on both orbits is $\mathbb{Z}$; restrictions are the identity and the transfer from C_2/e to C_2/C_2 is multiplication by two. These conventions agree with [1, Example 2.5.2].

For a C_2-space Y, write $Y_+ = Y \amalg \{*\}$ for the space with a disjoint C_2-fixed basepoint, and let

$$\mathbb{X}_Y = \Sigma^\infty_{C_2} Y_+, \qquad \mathbb{E}_Y = H\underline{\mathbb{Z}} \wedge \mathbb{X}_Y.$$

We use the $\mathbb{Z}$-graded Bredon homology group at the C_2/C_2-level:

$$H_q^{C_2}(Y;\underline{\mathbb{Z}}) = \pi_q\big((\mathbb{E}_Y)^{C_2}\big). \tag{7.1}$$

May constructs the equivariant Eilenberg–Mac Lane spectrum $H\underline{\mathbb{Z}}$ in [4, Theorem XIII.4.1]. The identification of the represented theory with Bredon homology is developed in the same section; the convention (7.1) is also reviewed in [1, Section 2.1].

All spectrum-level smash products, categorical fixed points, geometric fixed points, and homotopy orbits in this section are understood in the derived sense. For a genuine C_2-spectrum E, write E_{naive} for the underlying naive C_2-spectrum obtained by restricting the indexing universe to trivial representations; in particular, the C_2-action is retained.

The maps u and v induce commuting maps of suspension spectra and hence of $\mathbb{E}_{\mathcal{C}_{\mathrm{or}}(M)}$. Every fixed-point, geometric-fixed-point, homotopy-orbit, and spectral-sequence construction below is functorial in these maps. Thus all groups and all homomorphisms in this section are R-linear.

7.1 Geometric fixed points

For $K = e, C_2$, restrict the genuine spectrum to K before applying Φ^K. The two standard properties of geometric fixed points give natural equivalences

$$\begin{aligned} \Phi^K \mathbb{E}_{\mathcal{C}_{\mathrm{or}}(M)} &\simeq \Phi^K H\underline{\mathbb{Z}} \wedge \Phi^K \Sigma^\infty_K \mathcal{C}_{\mathrm{or}}(M)_+ \\ &\simeq \Phi^K H\underline{\mathbb{Z}} \wedge \Sigma^\infty \big(\mathcal{C}_{\mathrm{or}}(M)^K\big)_+. \end{aligned} \tag{7.2}$$

The first equivalence uses strong symmetric monoidality [1, Proposition 2.2.3], and the second uses compatibility with suspension spectra [1, Proposition 2.2.2].

Put

$$A_i^K = \pi_i \Phi^K H\underline{\mathbb{Z}}.$$

The constant Mackey functor is a commutative Green functor and has finitely generated values. It therefore follows from [1, Proposition 2.5.1] that $\Phi^K H\underline{\mathbb{Z}}$ is a connective generalized Eilenberg–Mac Lane spectrum and that every A_i^K is a finitely generated abelian group. Consequently there is a noncanonical splitting

$$\Phi^K H\underline{\mathbb{Z}} \simeq \bigvee_{i\geq 0} \Sigma^i H A_i^K.$$

After choosing this splitting once and for all, it is natural in the space factor. Since u and v act only on that factor, the resulting decomposition is R-linear. Smash products preserve wedges, and stable homotopy groups commute with wedges. Connectivity implies that, for a fixed $q \geq 0$, only the summands with $0 \leq i \leq q$ contribute. Thus (7.2) gives a natural R-module isomorphism

$$\pi_q \Phi^K \mathbb{E}_{\mathcal{C}_{\mathrm{or}}(M)} \cong \bigoplus_{i=0}^{q} H_{q-i}(\mathcal{C}_{\mathrm{or}}(M)^K; A_i^K). \tag{7.3}$$

Proposition 7.1 (Geometric-fixed-point finite generation)**.** *For $K = e, C_2$ and every $q \geq 0$, the R-module*

$$\pi_q \Phi^K \mathbb{E}_{\mathcal{C}_{\mathrm{or}}(M)}$$

is finitely generated.

Proof. For $K = e$, integral ordinary homology is finitely generated over R by Proposition 6.7; for $K = C_2$, it is finitely generated by Proposition 6.6. Each A_i^K is a finitely generated abelian group, so Lemma 6.8 applies to every summand of (7.3). There are only finitely many summands for fixed q. □

7.2 Isotropy separation for C_2

We follow the isotropy-separation approach of [1, Section 2.4]. For C_2, we prove the required finite-generation statement directly, including R-linearity and convergence of the homotopy-orbit spectral sequence.

Lemma 7.2 (The C_2 isotropy-separation step)**.** *Let E be a connective genuine C_2-spectrum equipped with two commuting genuine endomorphisms u, v, inducing an action of $R = \mathbb{Z}[u, v]$ on the groups below. Suppose that*

$$\pi_q(E_{\mathrm{naive}}) \quad \text{and} \quad \pi_q(\Phi^{C_2} E)$$

are finitely generated R-modules for all $q \geq 0$. Then $\pi_q(E^{C_2})$ is a finitely generated R-module for every $q \geq 0$.

Proof. The isotropy-separation cofiber sequence of [1, Lemma 2.4.1] has the form

$$(EC_{2+} \wedge E)^{C_2} \longrightarrow E^{C_2} \longrightarrow \Phi^{C_2} E. \tag{7.4}$$

Here EC_2 is a free contractible C_2-CW complex. Choose a cofibrant model of E, so that $EC_{2+} \wedge E$ is good and free in the sense required by the finite-group Adams equivalence [11, Main Theorem 1.7]. That equivalence and the underlying equivalence $EC_{2+} \wedge E \to E$ give natural equivalences

$$(EC_{2+} \wedge E)^{C_2} \simeq \big((EC_{2+} \wedge E)_{\mathrm{naive}}\big)_{hC_2} \simeq (E_{\mathrm{naive}})_{hC_2}.$$

There is no adjoint-representation shift for the finite group C_2. The skeletal filtration of the homotopy-orbit construction gives a first-quadrant spectral sequence

$$E^2_{p,q} = H_p(C_2; \pi_q(E_{\mathrm{naive}})) \Longrightarrow \pi_{p+q}\big((EC_{2+} \wedge E)^{C_2}\big). \tag{7.5}$$

For $p > 0$, the periodic resolution of the trivial $\mathbb{Z}[C_2]$-module expresses the group homology terms as kernels modulo images of $1 - g$ and $1 + g$; for $p = 0$, the term is the coinvariant quotient $\pi_q(E_{\mathrm{naive}})/(1 - g)\pi_q(E_{\mathrm{naive}})$. These maps are R-linear because u and v commute with the retained C_2-action. Since R is Noetherian, all these subquotients are finitely generated, as are the subquotients on subsequent pages. In each bidegree, the first-quadrant condition permits only finitely many incoming or outgoing differentials, so the E^∞-term is attained on a finite page and is finitely generated.

Write F_s for the s-th stage of the skeletal filtration. The quotient F_s/F_{s-1} is a wedge of s-fold suspensions of the underlying connective spectrum. The long exact sequence therefore shows that $\pi_t F_{s-1} \to \pi_t F_s$ is an isomorphism for $s > t + 1$. Consequently, $\pi_t F_{t+1} \to \pi_t F_\infty$ is an isomorphism, and the spectral sequence is strongly convergent with a finite filtration in each total degree. The abutment is therefore an iterated extension of finitely generated R-modules and is finitely generated.

Finally, put $U = (EC_{2+} \wedge E)^{C_2}$ and $V = \Phi^{C_2}E$. The R-linear long exact sequence of (7.4) gives

$$\begin{aligned} 0 \longrightarrow\ & \operatorname{coker}(\pi_{q+1}V \longrightarrow \pi_q U) \longrightarrow \pi_q(E^{C_2}) \\ \longrightarrow\ & \ker(\pi_q V \longrightarrow \pi_{q-1}U) \longrightarrow 0. \end{aligned}$$

The left term is a quotient of the finitely generated module $\pi_q U$; the right term is a submodule of the finitely generated module $\pi_q V$, hence finitely generated by Noetherianity. Their extension $\pi_q(E^{C_2})$ is therefore finitely generated. This also covers $q = 0$, since $\pi_{-1}U = 0$. □

Proof of Theorem 1.1. Take

$$E = \mathbb{E}_{\mathcal{C}_{\mathrm{or}}(M)} = H\underline{\mathbb{Z}} \wedge \Sigma^\infty_{C_2}\mathcal{C}_{\mathrm{or}}(M)_+.$$

This spectrum is connective: a nonnegative-dimensional equivariant CW approximation of $\mathcal{C}_{\mathrm{or}}(M)$ gives a filtration whose successive quotients are nonnegative suspensions of $H\underline{\mathbb{Z}} \wedge C_2/H_+$. At the trivial subgroup,

$$\pi_q(E_{\mathrm{naive}}) \cong H_q(\mathcal{C}_{\mathrm{or}}(M)^e; \mathbb{Z}),$$

which is finitely generated over R by Proposition 6.7. At C_2, Proposition 7.1 gives finite generation of $\pi_q(\Phi^{C_2}E)$. Applying Lemma 7.2 and the definition (7.1) yields

$$H^{C_2}_q(\mathcal{C}_{\mathrm{or}}(M); \underline{\mathbb{Z}}) = \pi_q(E^{C_2})$$

as a finitely generated R-module for every q. □

7.3 A single lift sector

Let

$$\mathcal{C}^\epsilon_{\mathrm{or}}(M) = \coprod_{n \geq 2} C^{+,\epsilon}_n(M), \qquad \epsilon \in \mathbb{Z}/2.$$

The maps u and v^2 preserve each sector. Put

$$R_0 = \mathbb{Z}[u, v^2].$$

Proof of Corollary 1.2. As an R_0-module,

$$R = R_0 \oplus vR_0$$

is finite free. The main theorem therefore implies that

$$H_d^{C_2}(\mathcal{C}_{\mathrm{or}}(M); \underline{\mathbb{Z}})$$

is finitely generated over R_0. The decomposition of the total family into its two lift sectors is a decomposition by open-and-closed invariant subspaces, so Bredon homology splits as

$$H_d^{C_2}(\mathcal{C}_{\mathrm{or}}(M); \underline{\mathbb{Z}}) \cong H_d^{C_2}(\mathcal{C}_{\mathrm{or}}^0(M); \underline{\mathbb{Z}}) \oplus H_d^{C_2}(\mathcal{C}_{\mathrm{or}}^1(M); \underline{\mathbb{Z}})$$

as R_0-modules. Each summand is the image of an R_0-linear projection and is therefore finitely generated. □

Remark 7.3. The conclusion is finite generation over the stated polynomial rings. It does not assert eventual isomorphisms under either stabilization separately: v interchanges the lift sectors, while the degree-zero calculation below exhibits nonzero classes annihilated by u^2.

8 Degree-zero Bredon homology

We compute degree zero directly from the Bredon cellular chain complex. This also verifies that the relations in (1.3) are complete.

Lemma 8.1 (Degree zero for a connected C_2-space)**.** *Let Y be a C_2-space of C_2-CW type. Suppose that Y is path connected and that*

$$\pi_0(Y^{C_2}) = \{F_1, \dots, F_c\}$$

is nonempty and finite. If z_i denotes the class of F_i, then

$$H_0^{C_2}(Y; \underline{\mathbb{Z}}) \cong \frac{\mathbb{Z}\langle z_1, \dots, z_c\rangle}{\langle 2(z_i - z_j) : 1 \le i, j \le c\rangle} \cong \mathbb{Z} \oplus (\mathbb{Z}/2)^{c-1}. \tag{8.1}$$

The construction is natural: an equivariant map sends z_i to the class of the fixed component containing its image.

Proof. The Bredon cellular chain complex is the coend of the fixed-point cellular coefficient system with the covariant orbit-category part of the Mackey functor; see [1, Section 2.1]. A coend is a colimit in abelian groups and therefore commutes with cokernels. Since degree-zero homology is the cokernel of the first cellular differential, we may first pass to ordinary degree-zero homology of the underlying and fixed-point cellular chain complexes and then impose the coend relations. For C_2, the orbit category is generated by the automorphism of C_2/e and the projection $C_2/e \to C_2/C_2$. The constant Mackey functor identifies the first automorphism with the identity and sends the transfer associated to the projection to multiplication by two.

The hypotheses give one underlying generator y and fixed-point generators $z_1, \dots, z_c$. The automorphism contributes only the identity relation, while the coend relation for the projection is

$$y = 2z_i \qquad (1 \le i \le c).$$

These orbit-category generators give all coend relations in degree zero. Eliminating y gives $2(z_i - z_j) = 0$ for all i, j. In the basis $z_1, z_2 - z_1, \dots, z_c - z_1$, the relations are precisely $2(z_i - z_1) = 0$ for $i \ge 2$, which gives the direct-sum decomposition in (8.1). This is the same cellular calculation used in the proof of [1, Proposition 4.2.1], here applied to one connected underlying component. Naturality follows from functoriality of the cellular coend. □

For $n \geq 2$ and $\epsilon \in \mathbb{Z}/2$, the underlying space of $C_n^{+,\epsilon}(M)$ is path connected. Its fixed points are nonempty: for $\epsilon = 0$ take $(a, b) = (n, 0)$, and for $\epsilon = 1$ take $(a, b) = (n-2, 1)$. Choose a standard equivariant CW approximation; it is a weak equivalence on the underlying and fixed-point spaces, so it preserves Bredon homology and the component data. Thus Lemma 8.1 applies to every fixed-arity summand $C_n^{+,\epsilon}(M)$.

For $a \geq 2$, let $z_{a,b}$ denote the class of the connected branch $D_{a,b}$. For $a = 0, 1$, let $z_{a,b}^+$ and $z_{a,b}^-$ denote the two component classes determined by the sign convention of Section 5. The indices always satisfy $a + 2b \geq 2$, and the ambient lift is $\bar{b}$.

Proposition 8.2 (Stabilization on component classes)**.** *The R-action on the fixed-component classes is*

$$\begin{aligned}
&uz_{a,b} = z_{a+1,b} && (a \geq 2), && vz_{a,b} = z_{a,b+1} && (a \geq 2),\\
&uz_{0,b}^{\pm} = z_{1,b}^{\pm}, && vz_{0,b}^{\pm} = z_{0,b+1}^{\pm} && (b \geq 1),\\
&uz_{1,b}^{+} = uz_{1,b}^{-} = z_{2,b}, && vz_{1,b}^{\pm} = z_{1,b+1}^{\pm} && (b \geq 1).
\end{aligned}$$

Proof. This is the map on path components in Propositions 5.2 and 5.3. In particular, λ_F sends the two signs to the two orientations inside the connected space $C_2^+(F)$; hence both components of $D_{1,b}$ land in the single component of $D_{2,b}$. □

Set

$$x = z_{2,0}, \qquad p = z_{0,1}^+, \qquad \delta = z_{0,1}^- - z_{0,1}^+. \tag{8.2}$$

Their bidegrees are

$$|x| = (2, 0), \qquad |p| = |\delta| = (2, 1).$$

Theorem 8.3 (Explicit degree-zero presentation)**.** *Let $R = \mathbb{Z}[u, v]$, with $|u| = (1, 0)$ and $|v| = (2, 1)$. There is an isomorphism of bigraded R-modules*

$$H_0^{C_2}(\mathcal{C}_{\mathrm{or}}(M); \underline{\mathbb{Z}}) \cong \frac{R\langle x, p\rangle}{(vx - u^2 p,\ 2(vp - u^2 x))} \oplus \frac{R}{(2, u^2)}\langle\delta\rangle. \tag{8.3}$$

Equivalently, it is generated by x, p, δ subject to

$$vx = u^2 p, \qquad 2(vp - u^2 x) = 0, \qquad 2\delta = 0, \qquad u^2\delta = 0. \tag{8.4}$$

Proof. We first verify the relations. By Proposition 8.2, both vx and $u^2 p$ are the class $z_{2,1}$, so they are equal. The two terms in

$$vp - u^2 x = z_{0,2}^+ - z_{4,0}$$

lie in distinct fixed components of $C_4^{+,0}(M)$, whose underlying space is connected. Their difference is killed by two by Lemma 8.1. Likewise $2\delta = 0$. Finally, u^2 sends both $z_{0,1}^+$ and $z_{0,1}^-$ to $z_{2,1}$, so $u^2\delta = 0$. Thus (8.4) defines a homomorphism from the presented module to Bredon homology.

It is surjective. Indeed, every fixed-component class has one of the following expressions:

$$z_{a,b} = u^{a-2}v^b x \qquad (a \geq 2), \tag{8.5}$$

$$z_{0,b}^+ = v^{b-1}p, \qquad z_{0,b}^- = v^{b-1}(p + \delta) \qquad (b \geq 1), \tag{8.6}$$

$$z_{1,b}^+ = uv^{b-1}p, \qquad z_{1,b}^- = uv^{b-1}(p + \delta) \qquad (b \geq 1). \tag{8.7}$$

To prove that no relation is missing, let P be the module presented by (8.4) and write

$$\tau = vp - u^2 x \in P.$$

Then $2\tau = 0$, $2\delta = 0$, and the relation $vx = u^2 p$ implies

$$v^2 x - u^4 x = u^2 \tau. \tag{8.8}$$

Assign to every fixed-component symbol the element of P given by (8.5)–(8.7). We check that the defining degree-zero relations from Lemma 8.1 hold.

Two allowed orbit types in the same arity and lift have b-coordinates of the same parity. For $a \geq 6$, adjacent orbit types whose fixed-point counts are both at least two are related by

$$(a, b) \longleftrightarrow (a - 4, b + 2).$$

The difference of the assigned elements is, up to sign,

$$u^{a-6} v^b (v^2 x - u^4 x) = u^{a-4} v^b \tau,$$

which is killed by two. The transition between the $a = 4$ and $a = 0$ lines has difference $v^b \tau$; the transition between the $a = 5$ and $a = 1$ lines has difference $uv^b \tau$. Finally, changing $+$ to $-$ adds a monomial multiple of δ, also killed by two. Iterating these elementary changes shows that twice the difference of any two fixed-component classes in the same summand $C_n^{+,\epsilon}(M)$ is zero in P.

The assignment therefore descends, by Lemma 8.1, to a homomorphism

$$H_0^{C_2}(\mathcal{C}_{\mathrm{or}}(M); \underline{\mathbb{Z}}) \longrightarrow P.$$

It respects u and v: the only nonformal check is the map from $a = 1$ to $a = 2$, where $vx = u^2 p$ and $u^2 \delta = 0$ give exactly Proposition 8.2. The two homomorphisms are inverse on x, p, δ, and hence are inverse everywhere. This proves the presentation. The direct-sum form in (8.3) follows because the relations involving δ do not mix it with x, p. □

Corollary 8.4 (Low-arity groups)**.** *The homogeneous groups in arities two through four include*

$$\begin{aligned}
H_0^{C_2}(C_2^{+,0}(M); \underline{\mathbb{Z}}) &\cong \mathbb{Z}\langle x \rangle, \\
H_0^{C_2}(C_2^{+,1}(M); \underline{\mathbb{Z}}) &\cong \mathbb{Z}\langle p \rangle \oplus \mathbb{Z}/2\langle \delta \rangle, \\
H_0^{C_2}(C_3^{+,1}(M); \underline{\mathbb{Z}}) &\cong \mathbb{Z}\langle up \rangle \oplus \mathbb{Z}/2\langle u\delta \rangle, \\
H_0^{C_2}(C_4^{+,1}(M); \underline{\mathbb{Z}}) &\cong \mathbb{Z}\langle vx = u^2 p \rangle, \\
H_0^{C_2}(C_4^{+,0}(M); \underline{\mathbb{Z}}) &\cong \mathbb{Z} \oplus (\mathbb{Z}/2)^2.
\end{aligned}$$

In the last line the two independent torsion classes may be represented by

$$vp - u^2 x \qquad \text{and} \qquad v\delta.$$

Proof. This follows either by taking the indicated homogeneous pieces of (8.3) or by counting the fixed components in Example 3.5 and applying Lemma 8.1. In arity four and lift zero there are three fixed components, so the two displayed order-two differences are independent. □

Comparison with unordered configurations. For the reflection action on $M = \mathbb{R}^r \oplus \sigma$, $r \geq 2$, fixed-point stabilization induces an injection

$$H_0^{C_2}(C_n(M); \underline{\mathbb{Z}}) \longrightarrow H_0^{C_2}(C_{n+1}(M); \underline{\mathbb{Z}}) \qquad (n \geq 2).$$

Indeed, the unordered fixed-point branches $I_{n-2b,b} \cong C_{n-2b}(F) \times C_b(B)$ are connected. Writing w_b for their component classes, Lemma 8.1 gives the free generator w_0 and independent order-two

generators $w_b - w_0$, for $1 \leq b \leq \lfloor n/2 \rfloor$. Stabilization preserves b, and hence sends these generators to the corresponding independent generators in the target.

For the oriented family, the direct summand $R/(2, u^2)\langle\delta\rangle$ in Theorem 8.3 gives

$$v^k\delta \neq 0, \qquad uv^k\delta \neq 0, \qquad u^2v^k\delta = 0 \qquad (k \geq 0),$$

and both nonzero classes have order two. Geometrically, v preserves the two orientation components on the branches with $a = 0, 1$. The first application of u takes $a = 0$ to $a = 1$ and preserves these components; the second takes $a = 1$ to $a = 2$, where both images lie in the same path component. The nonzero kernel class $uv^k\delta$ lies in arity $3 + 2k$ and lift $\overline{k+1}$. Thus fixed-point stabilization fails to be injective in arbitrarily large arities in each lift sector.

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